

Tongue2Speech: Real-Time Speech Synthesis from Tongue Ultrasound Videos via Spatiotemporal Transformers

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Abstract

Ultrasound-to-Speech aims to synthesize intelligible speech from tongue ultrasound video. Prior approaches rely mainly on purely convolutional or recurrent architectures. We present *Tongue2Speech*, a lightweight framework combining explicit 3D spatio-temporal encoding with a multi-layer Transformer to model long-range articulatory dynamics. Spectral distortion metrics (such as MSE) commonly used in prior work are weak indicators of lexical accuracy when used in isolation. Therefore, we evaluate performance using word error rate (WER). On the English TaL corpus, *Tongue2Speech* achieves 15.93% WER in single-speaker and 31.64% in multi-speaker settings, outperforming six strong baselines. Speaker-specific fine-tuning further improves intelligibility for unseen users. These results highlight tongue-only ultrasound as a promising pathway toward practical silent speech synthesis. Audio samples are available at: <https://tongue2speech.github.io/>

Index Terms: Ultrasound-to-Speech, Silent Speech Interfaces, Ultrasound Tongue Imaging

1. Introduction

Human speech arises from coordinated physiological processes in which airflow from the lungs excites the vocal folds and the vocal tract shapes the resulting sound through movements of the tongue, lips, jaw, and velum. As the most dynamic and acoustically influential articulator, the tongue governs vowel formants and many consonantal contrasts through its deformable surface [1, 2, 3], making its motion highly informative of the intended speech signal. The task of Ultrasound-to-Speech (U2S) aims to reconstruct intelligible speech from real-time tongue imaging, enabling Silent Speech Interfaces (SSI) without an acoustic signal [3]. Such systems are vital for individuals with preserved tongue movements but impaired phonation, such as laryngectomy patients [4] or those with neurological disorders [5, 6] that compromise laryngeal or respiratory control.

Existing SSI techniques present several challenges: electromagnetic articulography requires intraoral sensors, real-time MRI is costly and temporally constrained, and surface electromyography measures indirect muscle activity rather than articulator geometry [7, 8, 9]. In contrast, ultrasound directly captures tongue surface geometry, providing access to lingual gestures that are otherwise invisible [10, 3]. The development of cost-effective ultrasound hardware [11] and the current availability of portable probes¹ further enhances its viability for practical deployment. Consequently, among sensing modalities,

Ultrasound Tongue Imaging (UTI) offers a favorable trade-off between physiological fidelity and practical deployability.

Although U2S research has made steady progress, several important limitations remain. First, modeling long-range coarticulation is critical for accurate speech reconstruction from tongue-only ultrasound videos. To address this, most existing methods rely on convolutional encoders combined with recurrent LSTM-based temporal modeling [12, 13]. However, more recent advances in self-attention architectures remain largely underexplored. Second, evaluation is often conducted on relatively small speaker sets and relies on spectral distortion metrics such as mean squared error (MSE) or mel-cepstral distortion (MCD) [14], which provide limited insight into perceptual intelligibility. Large-scale multi-speaker WER-based assessment remains scarce, particularly for tongue-only systems [15, 16, 17]. Moreover, UTI is inherently speaker-dependent: anatomical differences, probe placement variation, and individual articulatory strategies produce substantial geometric variability across speakers. Hence, zero-shot cross-speaker generalization remains extremely challenging.

To this end, we introduce *Tongue2Speech*, a lightweight architecture that maps ultrasound tongue imaging to mel-spectrograms. We replace traditional LSTM or purely convolutional backbones with a 3D spatio-temporal front-end and a multi-layer Transformer encoder to capture long-range dynamics. Moving beyond standard spectral distortion metrics, we prioritize ASR-based word error rate (WER) as the principal measure of intelligibility. Our core contributions are:

- We propose *Tongue2Speech*, a U2S architecture that combines 3D convolutional encoding with a self-attention Transformer to model spatio-temporal dynamics and long-range coarticulation. The lightweight, non-autoregressive design uses only 19.17M parameters (6.25M excluding the vocoder), enabling real-time applications.
- We present an extensive investigation into U2S, providing comprehensive single- and multi-speaker performance across six baselines. We report both MSE and WER, with WER serving as the primary measure of usability.
- Our method achieves high-fidelity reconstruction with WERs of 16% in single-speaker and 31% in multi-speaker evaluations, demonstrating that *natural sounding* and *intelligible* U2S is achievable with tongue-only input. We show that zero-shot multi-speaker performance remains challenging, but speaker-specific fine tuning significantly restores intelligibility for unseen users.

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¹<https://www.butterflynetwork.com/iq3>

2. Related Work

Ultrasound imaging of tongue motion has been used in phonetic research since the 1960s, with systematic acquisition protocols established by Stone and colleagues in the 1980s and 1990s [7]. The U2S problem was first demonstrated by Denby and Stone [18], who mapped reconstructed tongue contours to vocal tract filter parameters using a multilayer perceptron and synthesized intelligible speech. A subsequent study [19] further evaluated spectral parameter prediction from ultrasound. The broader SSI framework was later formalized in [3], defining sensing modalities and application scenarios.

Modern U2S systems have evolved substantially in architectural design. Early approaches employed fully connected DNNs for articulatory-to-acoustic mapping [12]. These were followed by convolutional encoders combined with recurrent temporal modeling (typically LSTMs), predicting mel spectrograms rendered with neural vocoders [13, 20]. Subsequent work incorporated transfer learning from Tacotron2-style TTS systems [17], multi-task learning [21], speaker embeddings [22], ConvLSTM hybrids, data augmentation [23], and geometric normalization using Spatial Transformer Networks [24]. More recently, conformer-based architectures introduced attention mechanisms into U2S modeling [14]. The release of the Tongue and Lips (TaL) corpus [25] enabled larger-scale English U2S research. Multi-modal systems such as TaLNet combine ultrasound and lip video streams [15], and co-learning frameworks incorporate auxiliary audio supervision [16].

Several studies have explored ultrasound-to-text silent speech recognition, demonstrating that tongue motion provides complementary phonetic information. Landmark-based approaches using DeepLabCut improved multi-speaker recognition performance [26, 27]. UTI-derived articulatory features have also enhanced ASR for atypical speech [28], and ultrasound has been used in multimodal speech enhancement through knowledge distillation [10].

Although prior work acknowledges session and speaker-level variability due to anatomical differences and probe placement in ultrasound imaging, systematic evaluation of large-scale multi-speaker generalization using intelligibility-based metrics remains limited, and robust zero-shot generalization remains challenging. Moreover, speaker personalization has received relatively little attention in U2S synthesis. In this work, we therefore investigate fine-tuning a multi-speaker model trained on a diverse speaker pool for a target speaker, showing that efficient personalization can restore intelligible speech synthesis while highlighting the need for stronger zero-shot U2S models.

3. Dataset and Preprocessing

We conduct experiments on the TaL corpus [25], a synchronized multi-speaker dataset of ultrasound tongue imaging, audio, and lip videos. The corpus comprises two subsets: **TaL1**, containing six sessions from a single native English male speaker, and **TaL80**, consisting of single-session recordings from 81 native English speakers, totaling approximately 24 hours of synchronized ultrasound, audio, and video data. Ultrasound was recorded at ~ 80 fps with a 92° field of view, producing raw frames of size 64×842 (scanlines $\times$ radial samples). Audio was captured at 48kHz (16-bit). We extract 80-dimensional Mel-spectrograms (window/FFT size 2048, hop 512) as acoustic targets.

Raw ultrasound frames are provided as scanline data ar-

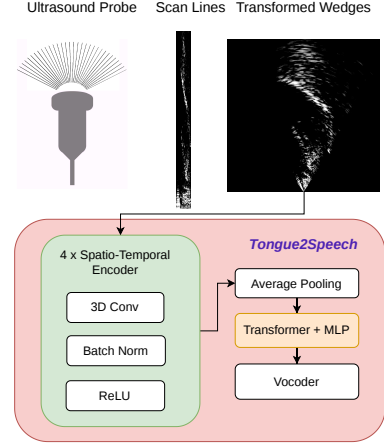


Figure 1: *Tongue2Speech*. Top: Raw scanlines from a portable ultrasound probe are transformed into a wedge representation. Bottom: Wedge frames are encoded by four spatio-temporal convolutional blocks, spatially pooled, and processed by a six-layer Transformer before synthesis via HiFi-GAN vocoder.

ranged in a (S, D, T) tensor, where $S = 64$ denotes scanlines, $D = 842$ radial samples, and T the number of frames. Because the data are acquired in polar coordinates, we transform the scanlines into a Cartesian wedge representation to better reflect tongue surface geometry. Assuming an angular sweep between θ_{start} and θ_{end} , the angular step is defined as $\Delta\theta = (\theta_{\text{end}} - \theta_{\text{start}}) / (S - 1)$, where S denotes the number of scanlines. Let Δd denote the radial resolution and $R_{\text{max}} = (D - 1)\Delta d$ the maximum depth, with D radial samples. For each Cartesian pixel (x, y) , polar coordinates are computed as $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan 2(y, x)$, yielding scanline indices $r_{\text{idx}} = r / \Delta d$ and $s_{\text{idx}} = (\theta - \theta_{\text{start}}) / \Delta\theta$.

We implement this mapping using GPU-accelerated bilinear grid sampling. A sampling grid and validity mask are pre-computed to exclude coordinates outside the probe’s field of view ($r > R_{\text{max}}$ or $\theta \notin [\theta_{\text{start}}, \theta_{\text{end}}]$). The resulting Cartesian wedge representation has spatial resolution $(64, 64)$ per frame and is used as input to the model.

4. Tongue2Speech

Tongue2Speech comprises three stages: a spatio-temporal convolutional encoder, a self-attention Transformer backbone, and a neural vocoder. Ultrasound sequences in wedge-transformed Cartesian geometry are processed by four stacked 3D convolutional blocks (Conv3D + BatchNorm + ReLU), operating jointly across spatial and short-term temporal dimensions to capture local articulatory motion. Spatial dimensions are aggregated via adaptive average pooling, producing a per-frame latent embedding of dimension 256 while preserving temporal resolution. The resulting latent sequence is processed by an encoder-only Transformer with six layers, eight attention heads per layer, and a feed-forward hidden dimension of 1024. Sinusoidal positional encodings preserve temporal order; each layer consists of multi-head self-attention followed by a position-wise feed-forward network with GELU activation and dropout ($p=0.1$). The final hidden states are projected to the acoustic domain using a two-layer MLP, yielding 80-dimensional mel-

Table 1: Recognition performance of baselines and the proposed Tongue2Speech on TaL1 (single speaker).

Session	Baselines				Model Variants (Ours)								Proposed	
	Conformer-U2S		STN-CNN		2D-CNN+BiLSTM		3D-CNN		3D-CNN+BiLSTM		3D-CNN+BiLSTM+Skip		Tongue2Speech (Ours)	
	WER	MSE	WER	MSE	WER	MSE	WER	MSE	WER	MSE	WER	MSE	WER	MSE
Day 2	100.00	1.815	100.00	1.833	62.86	1.376	100.00	3.120	38.12	0.810	21.55	0.746	16.45	0.635
Day 3	98.47	1.408	98.15	1.402	58.63	1.305	98.47	2.987	30.82	0.623	15.61	0.474	14.96	0.525
Day 4	100.00	1.980	100.00	2.082	56.15	1.972	100.00	3.275	35.75	1.116	28.12	1.055	20.79	0.657
Day 5	98.67	1.970	98.07	1.877	62.86	1.291	98.67	3.289	33.66	1.002	24.24	0.854	11.76	0.587
Day 6	99.28	1.923	99.28	2.157	60.94	1.213	99.28	3.122	42.84	0.981	30.24	0.850	15.93	0.579
Overall	99.26	1.819	99.08	1.877	60.29	1.417	99.26	3.155	36.46	0.906	24.15	0.793	15.93	0.594

spectrogram frames. Predicted mel-spectrograms are converted to waveform audio using a HiFi-GAN vocoder [29] configured to match the mel extraction parameters in Section 3. Separate vocoders are trained for the single-speaker and multi-speaker regimes. *Tongue2Speech* has 19.17M parameters in total, with 6.25M in the convolutional front-end and Transformer backbone and 13.92M in the neural vocoder.

5. Baselines

We evaluate *Tongue2Speech* against six baselines: Conformer-U2S [14], STN-CNN [24], 3D-CNN, 3D-CNN+BiLSTM, 3D-CNN+BiLSTM-Skip, and 2D-CNN+BiLSTM. The baselines span the major design axes of U2S mapping: input dimensionality (2D vs. 3D), temporal modeling (no explicit temporal aggregation vs. BiLSTM), and multi-scale feature aggregation. Since the source code for Conformer-U2S and STN-CNN is not publicly available, we re-implement both methods. The remaining four baselines are purpose-built architectural variants designed to isolate specific modeling factors. Unless stated otherwise, all baselines share the same mel extraction pipeline, vocoder, optimizer, and data splits. Conformer-U2S and STN-CNN are trained under their original MSE objectives, whereas *Tongue2Speech* and all proposed variants use L_1 loss on mel spectrograms. For MSE evaluation, we compute log-mel spectrograms by clamping values to a minimum of 10^{-7} before applying the natural logarithm. For the re-implemented baselines [14, 24], the generated samples qualitatively match the originals (both the spectrograms and speech); however, MSE values may differ due to variations in normalization procedures (see webpage for samples).

Conformer-U2S [14] processes resized scanline frames (64×128) through a single Conformer block (encoder 256 dimensions, 32 attention heads, convolution kernel size 31) followed by two Bi-LSTM layers, combining multi-head self-attention, depthwise separable convolutions, and recurrent temporal modeling. **STN-CNN** [24] prepends a Spatial Transformer Network module to a CNN regression backbone, which learns an affine transformation of input frames to compensate for probe misalignment and inter-speaker geometric variability.

3D-CNN captures joint spatio-temporal features through four 3D convolutional blocks (Conv3D + BatchNorm + ReLU + Dropout, $p=0.1$) with $3 \times 3 \times 3$ kernels and spatial-only downsampling, followed by a linear mel projection. **3D-CNN+BiLSTM** augments three 3D convolutional blocks (Conv3D + BatchNorm + MaxPool3D) with a two-layer bidirectional LSTM (hidden 256 dimensions) before mel projection. **3D-CNN+BiLSTM-Skip** uses a three-stage 3D convolutional encoder ($3 \times 3 \times 3$ kernels, BatchNorm, ReLU) with

$1 \times 2 \times 2$ MaxPool3d layers that downsample spatial resolution by a factor of 8 while preserving temporal length. A multi-scale skip-connection mechanism projects intermediate feature maps via $1 \times 1 \times 1$ convolutions, trilinearly interpolates them to a unified 8×8 resolution, and concatenates them through a fusion layer. The features are passed to a two-layer, 256-unit BiLSTM followed by an MLP. **2D-CNN+BiLSTM** processes each frame independently through four 2D convolutional blocks (Conv2D + BatchNorm + ReLU + MaxPool2D), omitting max-pooling in the final block. Spatial features are collapsed to 1×1 via adaptive average pooling and passed to a three-layer BiLSTM (hidden 256 dimensions), with a linear output projection.

6. Results

We evaluate *Tongue2Speech* and six baselines across single-speaker, multi-speaker, and personalized settings. Synthesized speech is evaluated using OpenAI Whisper Medium ASR [30].

6.1. Single-Speaker Experiments

Table 1 reports single-speaker results on the TaL1 corpus [25]. Day 1 is excluded due to synchronization issues.² The proposed *Tongue2Speech* substantially outperforms all baselines in terms of intelligibility. It achieves the lowest overall WER of 15.93%, improving over the strongest baseline (3D-CNN+BiLSTM w/ Skip, 24.15%) by more than 8 absolute points while also obtaining the lowest overall MSE (0.594). These results indicate that explicit spatio-temporal modeling combined with Transformer-based temporal aggregation significantly improves reconstruction quality from tongue ultrasound. In contrast, Conformer-U2S and STN-CNN produce very high MSE values (1.8–1.9) and near-saturated WER ($\sim 99\%$), indicating effectively unintelligible output. These results show that these architectures fail to learn a meaningful articulatory-to-acoustic mapping under the current experimental setup.

Across architectural variants, long-range temporal aggregation and 3D spatio-temporal encoding emerge as key factors for intelligible speech reconstruction. Systems relying solely on convolution or weak temporal aggregation (3D-CNN, Conformer-U2S, STN-CNN) produce near-saturated WER, whereas BiLSTM and Transformer-based models substantially reduce error. Although Conformer-U2S includes attention, it applies only a single Conformer block to resized 2D frames and is trained with an MSE objective. In contrast, *Tongue2Speech* uses a multi-layer Transformer operating on motion-aware embeddings learned by a 3D spatio-temporal front end and is

²https://ultrasuite.github.io/data/tal_corpus/#synchronisation-of-data-streams

Table 2: *Multi-speaker evaluation on TaL80. Results are reported for multi-speaker training and speaker-specific fine-tuning of Tongue2Speech on held-out speakers.*

Setting	Configuration / Speaker	WER	MSE
Multi-speaker	3D-CNN + BiLSTM	42.46	0.981
	3D-CNN + BiLSTM w/ Skip	33.23	0.863
	Tongue2Speech	31.64	0.851
Personalization	Speaker 78	33.00	0.860
	Speaker 79	46.07	1.013
	Speaker 80	27.62	0.852
	Speaker 81	37.87	0.914

trained with an L_1 mel loss. The 24-point WER gap between 2D-CNN+BiLSTM (60.29%) and 3D-CNN+BiLSTM (36.46%) further highlights the benefit of joint space-time modeling of articulatory motion. *Tongue2Speech* achieves the best overall intelligibility and performs particularly well on Days 2&5 (14.96%, 11.76%). Nonetheless, session-level variability of up to 9 absolute WER points persists across systems, reflecting sensitivity to probe placement and inter-session geometric variation.

6.2. Multi-Speaker Experiments

Multi-speaker experiments are conducted on the TaL80 corpus [25], comprising articulatory recordings from 81 speakers with anatomical and probe-placement variability. We evaluate the two strongest single-speaker baselines (3D-CNN+BiLSTM and 3D-CNN+BiLSTM w/ Skip) alongside *Tongue2Speech*.

A. Multi-Speaker Training. All 81 speakers are pooled and partitioned using a 95/5 train-test split at the utterance level. This setting evaluates the ability to learn speaker-robust articulatory representations under mixed-speaker supervision. Results are shown in Table 2 (Section A). Compared to the single-speaker setting, WER increases substantially across all systems, highlighting the difficulty of modeling cross-speaker articulatory variability. *Tongue2Speech* achieves the lowest WER (31.64%), outperforming 3D-CNN+BiLSTM (42.46%) and the skip variant (33.23%). Despite broadly comparable MSE values, intelligibility differences persist, indicating that attention-based aggregation over 3D motion embeddings improves robustness to speaker heterogeneity, though cross-speaker variability remains a limiting factor.

B. Personalization via Fine-Tuning. To assess adaptation to unseen speakers, *Tongue2Speech* is first trained on 77 speakers (1500 epochs), with four speakers (IDs 78–81) held out. For each held-out speaker, the pre-trained model is fine-tuned on 95% of that speaker’s data for 20 epochs and evaluated on the remaining 5%. Results are reported in Table 2 (Section B). After fine-tuning, WER ranges between 27.62–46.07% across speakers, with the best result observed for Speaker 80 (27.62%). These results indicate that while cross-speaker generalization remains challenging, limited speaker-specific adaptation substantially improves intelligibility without architectural modification. Together, these findings support personalization as a practical deployment strategy for tongue-only U2S.

6.3. Qualitative Evaluation

Figure 2 provides a qualitative comparison of mel-spectrograms generated by all systems against ground truth. Conformer-U2S [14] and STN-CNN [24] exhibit over-smoothed energy

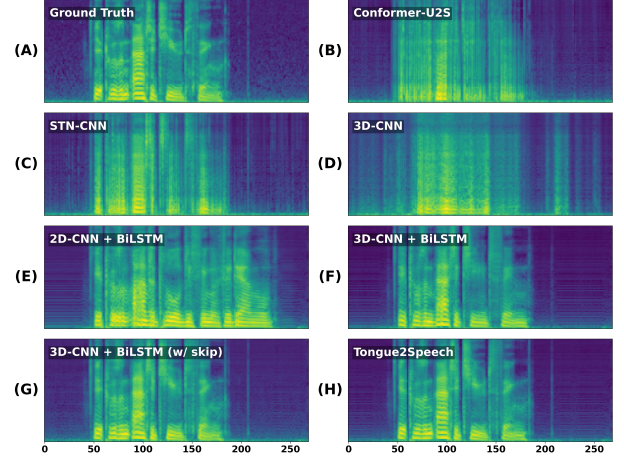


Figure 2: *Mel-spectrogram comparison between ground-truth speech and model outputs for a representative TaL1 utterance. (A) Ground truth. (B) Conformer-U2S. (C) STN-CNN. (D) 3D-CNN. (E) 2D-CNN+BiLSTM. (F) 3D-CNN+BiLSTM. (G) 3D-CNN+BiLSTM+Skip. (H) Tongue2Speech. Models with weak temporal aggregation exhibit spectral smearing and blurred structure, whereas Tongue2Speech more closely preserves harmonics, formant trajectories, and transient boundaries.*

distributions and blurred harmonic structure, with reduced formant contrast and weakened temporal transitions, consistent with their near-saturated WER. The 3D-CNN baseline shows limited temporal sharpness, while 2D-CNN+BiLSTM produces visible horizontal banding artifacts and reduced high-frequency detail, indicating insufficient modeling of joint spatial-temporal motion. Incorporating 3D convolution with BiLSTM substantially restores formant continuity and vertical transition structure, and the skip variant further sharpens low-frequency energy bands and improves temporal localization. *Tongue2Speech* most closely matches the ground-truth spectrogram, preserving harmonic stacks, formant trajectories, and transient boundaries with improved temporal consistency. These qualitative differences align with the quantitative WER results, suggesting that accurate modeling of articulatory motion dynamics directly translates to improved phonetic fidelity and intelligibility.

7. Conclusion

We presented *Tongue2Speech*, a lightweight tongue-only ultrasound-to-speech architecture that combines 3D spatio-temporal convolution with a Transformer encoder for articulatory-to-acoustic mapping. Across six baselines, ASR-based evaluation shows that explicit modeling of articulatory motion and long-range temporal dependencies substantially improves intelligibility over purely convolutional or weakly aggregated systems. Our results further demonstrate that spectral MSE does not reliably reflect lexical correctness, reinforcing the need for intelligibility-driven evaluation in U2S research. Overall, our results demonstrate that intelligible speech can be reconstructed from tongue ultrasound across single-speaker, multi-speaker, and personalized settings. Nevertheless, robust cross-speaker generalization and efficient personalization remain important open challenges and key directions for future research in ultrasound-to-speech synthesis.

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9. Generative AI Use Disclosure

Generative AI tools were used only to improve grammar, phrasing, and clarity of expression. All conceptual and scientific contributions originate from the authors.

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